

PUNJAB UNIVERSITY COLLEGE OF INFORMATION TECHNOLOGY

University of the Punjab

Course: Introduction to Data Science / Machine Learning **Total Marks: 85 | Time: 1.75 Hours**

Section A: Objective (25 Marks)

Q1. Multiple Choice Questions *This section consists of 25 MCQs covering topics including Linear Regression, Logistic Regression, Decision Trees, Gradient Descent variants, and Ensemble techniques.*

Section B: Subjective (60 Marks)

Answer all questions. Each question carries 11–12 marks.

Question 1: Linear Regression & Optimization (11 Marks)

1. Consider the following hypothesis equation: State whether this equation is considered **Linear** or **Non-linear**. Justify your answer based on the role of parameters (weights) versus features. **(3 Marks)**
2. Explain the conceptual reason behind your "Yes/No" choice above. How do we define linearity in the context of Regression models? **(3 Marks)**
3. In Machine Learning, we often use **Gradient Descent** to find optimal weights. Why do we prefer Gradient Descent over simple **Matrix Multiplications** (Normal Equation) to find approximate weights, especially as the dataset size increases? Discuss in terms of computational cost. **(5 Marks)**

Question 2: Logistic Regression & Decision Boundaries (12 Marks)

1. You are given a 2D graph where Class A is concentrated in the center (forming a circle) and Class B is distributed around it. Can **Logistic Regression** be used to separate these two classes? Briefly explain. **(3 Marks)**
2. Based on your answer, identify the shape of the decision boundary that would be required to separate these classes (e.g., Linear, Oval, Circular). **(2 Marks)**
3. **Graphing Task:** Sketch a representative graph showing Class 'X' and Class 'O' where one is enclosed by the other. Draw the appropriate decision boundary. **(4 Marks)**
4. Write the mathematical update equation for **either** Batch Gradient Descent or Stochastic Gradient Descent. Highlight one key difference between the two. **(3 Marks)**

Question 3: Decision Trees - Numerical Analysis (12 Marks)

1. **Case Study:** You are provided with a dataset (similar to the "Play Tennis" dataset).
 - Identify the **Target Class** in this scenario. **(3 Marks)**
 - Define what a "Target Class" represents in a supervised learning model. **(2 Marks)**
2. Using the provided table (Attributes: Outlook, Temp, Humidity, etc.), calculate the following to determine the root node:
 - **Entropy** of the system.
 - **Information Gain** for a specific attribute.
 - **Split Information** (if applicable). **(7 Marks)**

Question 4: Ensemble Methods & Random Forest (11 Marks)

1. Explain the theoretical framework of a **Random Forest**. How does it differ from a single Decision Tree? **(4 Marks)**
2. Define the following terms in the context of Ensemble Learning: **(7 Marks)**
 - Bootstrapping
 - Bagging (Bootstrap Aggregating)
 - Sampling With Replacement vs. Without Replacement

Question 5: Model Tuning & Pruning (11 Marks)

1. What is a **Hyperparameter**? How does it differ from a model parameter? **(3 Marks)**
2. Name and briefly describe **two** specific hyperparameters used in Decision Trees (e.g., related to depth or leaf nodes). **(3 Marks)**
3. To prevent overfitting in Decision Trees, we use **Pruning**. Explain the following three pruning techniques: **(5 Marks)**
 - Pre-Pruning
 - Reduced Error Pruning
 - Rule Post-Pruning

End of Paper